

Unit 6. Orthogonality, Projections, and Least Squares

1. Inner products and orthonormal bases

For real columns $x, y \in \mathbb{R}^n$, define the **inner product** and **Euclidean norm** by

$$x \cdot y = x^\top y = \sum_{i=1}^n x_i y_i, \quad \|x\| = \sqrt{x^\top x} = \sqrt{\sum_{i=1}^n x_i^2}.$$

The finite sum gives symmetry and bilinearity:

$$x^\top y = y^\top x, \quad (ax + by)^\top z = ax^\top z + by^\top z.$$

Linearity in the second argument follows by symmetry. Every x_i^2 is nonnegative, and their sum is zero exactly when all $x_i = 0$. Thus

$$\|x\| \geq 0, \quad \|x\| = 0 \iff x = 0, \quad \|cx\| = \sqrt{c^2 \sum_i x_i^2} = |c| \|x\|.$$

Vectors x, y are **orthogonal**, written $x \perp y$, if $x^\top y = 0$. Expanding the product gives

$$\|x + y\|^2 = \|x\|^2 + 2x^\top y + \|y\|^2.$$

In particular, $x \perp y$ implies the **Pythagorean identity** $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Theorem 6.1 (Schwarz and triangle inequalities). For $x, y \in \mathbb{R}^n$,

$$|x^\top y| \leq \|x\| \|y\|, \quad \|x + y\| \leq \|x\| + \|y\|.$$

Equality in the first holds exactly when $y = 0$ or $x = ty$ for some real t . Equality in the second holds exactly when $y = 0$ or $x = ty$ for some $t \geq 0$.

Proof. • For $y = 0$, both equalities hold. For $y \neq 0$, put $t = (x^\top y)/(y^\top y)$. Expansion yields

$$0 \leq \|x - ty\|^2 = \|x\|^2 - \frac{(x^\top y)^2}{\|y\|^2}.$$

Multiply by $\|y\|^2 > 0$ and take nonnegative square roots. Equality is equivalent to $\|x - ty\|^2 = 0$, hence to $x = ty$. Conversely, $x = ty$ gives $|x^\top y| = |t| \|y\|^2 = \|x\| \|y\|$.

- By the expansion above and Schwarz,

$$\|x + y\|^2 \leq \|x\|^2 + 2|x^\top y| + \|y\|^2 \leq (\|x\| + \|y\|)^2.$$

Taking square roots proves the triangle inequality. For $y \neq 0$, equality requires equality in both steps: $x = ty$ and $x^\top y \geq 0$. Since $x^\top y = t \|y\|^2$, this is exactly $t \geq 0$. Conversely, that condition gives $\|x + y\| = (t + 1) \|y\| = \|x\| + \|y\|$.

□

A list $(u_1, \dots, u_r)$ is **orthogonal** if $u_i^\top u_j = 0$ whenever $i \neq j$. It is **orthonormal** if, in addition, $u_i^\top u_i = 1$ for every i .

Proposition 6.2. Nonzero orthogonal vectors $u_1, \dots, u_r$ are independent. If x belongs to their span, then

$$x = \sum_{i=1}^r \frac{u_i^\top x}{u_i^\top u_i} u_i.$$

For an orthonormal list $(q_1, \dots, q_r)$, this becomes $x = \sum_i (q_i^\top x) q_i$.

- Proof.* • If $\sum_i c_i u_i = 0$, multiply by $u_j^\top$. All cross terms vanish, leaving $c_j \|u_j\|^2 = 0$. Since $u_j \neq 0$, $c_j = 0$ for each j .
- If $x = \sum_i c_i u_i$, the same multiplication gives $u_j^\top x = c_j u_j^\top u_j$, which determines the displayed coefficients. For an orthonormal list, the denominators are 1.

□

Theorem 6.3 (Gram–Schmidt). For an independent list $v_1, \dots, v_r \in \mathbb{R}^n$, $r \geq 1$, define successively

$$u_1 = v_1, \quad u_j = v_j - \sum_{i < j} \frac{u_i^\top v_j}{u_i^\top u_i} u_i \quad (2 \leq j \leq r).$$

The vectors $u_1, \dots, u_r$ are nonzero and orthogonal. For every j ,

$$\text{span}(u_1, \dots, u_j) = \text{span}(v_1, \dots, v_j).$$

Consequently, $q_j = u_j / \|u_j\|$ form an orthonormal basis of $\text{span}(v_1, \dots, v_r)$.

Proof. • For $j = 1$, $u_1 = v_1 \neq 0$, and the spans agree. Suppose the conclusions hold through $j - 1$. All denominators in the definition of u_j are positive.

- For $k < j$, orthogonality of the preceding vectors gives

$$u_k^\top u_j = u_k^\top v_j - \frac{u_k^\top v_j}{u_k^\top u_k} u_k^\top u_k = 0.$$

If $u_j = 0$, its definition puts v_j in the span of the preceding u_i , hence of $v_1, \dots, v_{j-1}$, a contradiction.

- The definition puts u_j in $\text{span}(v_1, \dots, v_j)$. The reverse equation

$$v_j = u_j + \sum_{i < j} \frac{u_i^\top v_j}{u_i^\top u_i} u_i$$

gives the reverse inclusion of spans, using the induction hypothesis for the preceding vectors. This completes the induction.

- Dividing each nonzero vector by its norm preserves the span and orthogonality, and gives norm 1. Independence follows from Proposition 6.2.

□

Every subspace $W \subseteq \mathbb{R}^n$ has an orthonormal basis: for $W \neq \{0\}$, apply Gram–Schmidt to a basis supplied by Unit 3, Theorem 3.3. For $W = \{0\}$, use the empty basis. An existing orthonormal list in W can be extended to an orthonormal basis of W : first extend it to a basis by Unit 3, Theorem 3.3, then apply Gram–Schmidt. Each existing vector is unchanged, because its inner products with its predecessors are zero and its norm is already 1.

Example 6.1. Find an orthonormal basis of the span of

$$a = (1, 1, 0, 1)^\top, \quad b = (1, -2, 0, 0)^\top, \quad c = (1, 0, -1, 2)^\top.$$

- $a^\top a = 3$ and $a^\top b = -1$, so

$$b' = b + \frac{1}{3}a = \frac{1}{3}(4, -5, 0, 1)^\top, \quad \|b'\|^2 = \frac{14}{3}.$$

- $a^\top c = 3$ and $b'^\top c = 2$, so

$$c' = c - a - \frac{3}{7}b' = \frac{1}{7}(-4, -2, -7, 6)^\top, \quad \|c'\|^2 = \frac{15}{7}.$$

The dot products $a^\top b', a^\top c', b'^\top c'$ are respectively $(4 - 5 + 1)/3, (-4 - 2 + 6)/7, (-16 + 10 + 6)/21$, all zero.

- All three outputs are nonzero. The identities $b = b' - a/3$, $c = c' + a + 3b'/7$ prove that they span the original space: the definitions give one span inclusion, and these reverse identities give the other. Orthogonality proves independence. After normalization an orthonormal basis is

$$\frac{(1, 1, 0, 1)^\top}{\sqrt{3}}, \quad \frac{(4, -5, 0, 1)^\top}{\sqrt{42}}, \quad \frac{(-4, -2, -7, 6)^\top}{\sqrt{105}}.$$

A real square matrix $Q : n \times n$ is **orthogonal** if $Q^\top Q = I_n$. Entry (i, j) of this product is $q_i^\top q_j$, so the condition says exactly that its columns are orthonormal. The one-sided inverse criterion of Unit 2 gives $Q^{-1} = Q^\top$, hence $QQ^\top = I_n$. Moreover,

$$\|Qx\|^2 = x^\top Q^\top Qx = \|x\|^2 \quad (x \in \mathbb{R}^n), \quad (\det Q)^2 = \det(Q^\top Q) = 1.$$

The second equality uses determinant multiplication and transpose invariance from Unit 5; thus $\det Q = 1$ or -1 .

Theorem 6.4 (Thin QR decomposition). Let $A = [a_1 \cdots a_p] : m \times p$ have independent columns, where $1 \leq p \leq m$. There is a unique factorization

$$A = QR, \quad Q^\top Q = I_p,$$

where $Q : m \times p$ and $R : p \times p$ is upper triangular with positive diagonal entries. Moreover, $R = Q^\top A$ and $\mathcal{C}(Q) = \mathcal{C}(A)$.

Proof. • Apply Gram–Schmidt to the columns of A . In terms of the normalized vectors, its recursion is

$$u_j = a_j - \sum_{i < j} (q_i^\top a_j) q_i, \quad r_{jj} = \|u_j\| > 0, \quad q_j = u_j / r_{jj}.$$

Theorem 6.3 ensures that each $u_j \neq 0$ and that the q_j are orthonormal. Set $r_{ij} = q_i^\top a_j$ for $i < j$, and $r_{ij} = 0$ for $i > j$. Then

$$a_j = \sum_{i < j} r_{ij} q_i + r_{jj} q_j \quad (1 \leq j \leq p),$$

which is precisely $A = QR$. Multiplication by $Q^\top$ gives $R = Q^\top A$. The prefix-span equality in Gram–Schmidt gives $\mathcal{C}(Q) = \mathcal{C}(A)$.

- For uniqueness, consider any such factorization. Its first column is $a_1 = r_{11}q_1$; hence $r_{11} = \|a_1\|$ and $q_1 = a_1 / \|a_1\|$. Suppose $q_1, \dots, q_{j-1}$ have been determined. Orthonormality in the j th column gives

$$r_{ij} = q_i^\top a_j \quad (i < j), \quad r_{jj}q_j = a_j - \sum_{i < j} (q_i^\top a_j) q_i = u_j.$$

Since $r_{jj} > 0$ and $\|q_j\| = 1$, necessarily $r_{jj} = \|u_j\|$ and $q_j = u_j / \|u_j\|$. Thus every column and every entry of the two factors is determined. □

For the vectors a, b, c in Example 6.1, the computed relations

$$b = -\frac{1}{3}a + b', \quad c = a + \frac{3}{7}b' + c'$$

give the QR factors directly:

$$[a \ b \ c] = QR, \quad Q = \begin{bmatrix} \frac{a}{\sqrt{3}} & \frac{3b'}{\sqrt{42}} & \frac{7c'}{\sqrt{105}} \end{bmatrix}, \quad R = \begin{pmatrix} \sqrt{3} & -1/\sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42}/3 & \sqrt{42}/7 \\ 0 & 0 & \sqrt{105}/7 \end{pmatrix}.$$

Each column of this product is the corresponding relation above (with $a = \sqrt{3}q_1$). Here Q has four rows and three columns; $Q^\top Q = I_3$, while $QQ^\top \neq I_4$, since its rank is at most three.

2. Orthogonal complements and projections

For a subspace $W \subseteq \mathbb{R}^n$, define its **orthogonal complement** (in $\mathbb{R}^n$) by

$$W^\perp = \{x \in \mathbb{R}^n : w^\top x = 0 \text{ for every } w \in W\}.$$

It contains zero, and $x, y \in W^\perp$ imply $w^\top(ax + by) = aw^\top x + bw^\top y = 0$ for all $w \in W$; thus it is a subspace. If $x \in W \cap W^\perp$, then $x^\top x = 0$, so $x = 0$. For any spanning list $w_1, \dots, w_r$ of W ,

$$x \in W^\perp \iff w_i^\top x = 0 \quad (1 \leq i \leq r).$$

The forward implication is the definition. Conversely, for $w = \sum_i c_i w_i$, $w^\top x = \sum_i c_i w_i^\top x = 0$.

Theorem 6.5. Let $W \subseteq \mathbb{R}^n$ be a subspace of dimension r . Then

$$\mathbb{R}^n = W \oplus W^\perp, \quad \dim W^\perp = n - r, \quad (W^\perp)^\perp = W.$$

Proof. • Extend an orthonormal basis $q_1, \dots, q_r$ of W to an orthonormal basis $q_1, \dots, q_n$ of $\mathbb{R}^n$, as proved above. Every $x \in \mathbb{R}^n$ has the unique expansion

$$x = \sum_{i=1}^n (q_i^\top x) q_i.$$

It belongs to $W^\perp$ exactly when its first r coefficients vanish. Thus $q_{r+1}, \dots, q_n$ are a basis of $W^\perp$.

- Splitting the expansion into its first r and last $n - r$ terms gives the sum $W + W^\perp = \mathbb{R}^n$. Its intersection is zero, as already proved, so the sum is direct and the dimensions follow.
- Symmetry of the inner product gives $W \subseteq (W^\perp)^\perp$. Applying the dimension formula twice gives $\dim(W^\perp)^\perp = n - (n - r) = r$. The equal-dimension subspace criterion of Unit 3 proves equality.

Empty lists and sums include $W = \{0\}$ and $W = \mathbb{R}^n$. □

Theorem 6.6 (Orthogonal projection). Let $W \subseteq \mathbb{R}^n$ be a subspace. For each $y \in \mathbb{R}^n$, there is exactly one $p \in W$ such that $y - p \in W^\perp$. For any orthonormal basis $(q_1, \dots, q_r)$ of W ,

$$p = \sum_{i=1}^r (q_i^\top y) q_i.$$

This vector is the **orthogonal projection** of y on W , denoted $P_W y$.

Proof. • The displayed vector belongs to W , and $q_j^\top(y - p) = q_j^\top y - q_j^\top p = 0$ for every j . The spanning-list test above gives $y - p \in W^\perp$.

- If p' also satisfies the two conditions, then $p - p' \in W$ and $p - p' = (y - p') - (y - p) \in W^\perp$. Their intersection is zero, so $p = p'$.

For $W = \{0\}$, the empty sum is $p = 0$, which has the same properties. □

Uniqueness shows that $P_W y$ is independent of the chosen orthonormal basis. The formula gives linearity:

$$P_W(ay + bz) = \sum_i q_i (q_i^\top (ay + bz)) = aP_W y + bP_W z.$$

Also $P_W y = y$ exactly when $y \in W$: one direction follows because the image lies in W ; in the other, $p = y$ has zero residual and therefore satisfies the defining conditions.

Theorem 6.7 (Closest point). Let $W \subseteq \mathbb{R}^n$ be a subspace and $p = P_W y$. For every $w \in W$,

$$\|y - w\|^2 = \|y - p\|^2 + \|p - w\|^2.$$

Consequently p is the unique minimizer of $\|y - w\|$ over $w \in W$. A vector $w \in W$ is that minimizer if and only if $y - w \in W^\perp$.

- Proof.* • Write $y - w = (y - p) + (p - w)$. The first summand belongs to $W^\perp$, the second to W ; hence their inner product is zero. The Pythagorean identity gives the displayed equality.
- The last term is nonnegative and is zero exactly when $w = p$. Thus the squared norm, and hence the norm, has its unique minimum at p .
 - If $y - w \in W^\perp$, projection uniqueness gives $w = p$. Conversely, every minimizer equals p , whose residual belongs to $W^\perp$.

□

Proposition 6.8 (Projection matrices). Let $W \subseteq \mathbb{R}^n$ be a subspace, and let $Q = [q_1 \ \cdots \ q_r]$ contain an orthonormal basis of W . The matrix of P_W in standard coordinates is

$$P = QQ^\top, \quad P^\top = P, \quad P^2 = P, \quad \mathcal{C}(P) = W, \quad N(P) = W^\perp.$$

Moreover, $I_n - P$ is the matrix of $P_{W^\perp}$. A real square matrix is an orthogonal projection matrix if and only if it is symmetric and idempotent.

Proof. • Matrix multiplication gives $QQ^\top y = \sum_i q_i(q_i^\top y) = P_W y$, so the matrices agree. Since $Q^\top Q = I_r$,

$$(QQ^\top)^\top = QQ^\top, \quad (QQ^\top)^2 = Q(Q^\top Q)Q^\top = QQ^\top.$$

- Every Py belongs to W , and every $w \in W$ satisfies $Pw = w$, proving $\mathcal{C}(P) = W$. If $y \in W^\perp$, all coefficients $q_i^\top y$ vanish, so $Py = 0$. Conversely, $Py = 0$ gives $y = y - Py \in W^\perp$.
- For every y , $(I_n - P)y \in W^\perp$ and its residual Py lies in $W = (W^\perp)^\perp$. Uniqueness of projection onto $W^\perp$ gives $(I_n - P)y = P_{W^\perp} y$ for every y .
- Conversely, suppose $H^\top = H$ and $H^2 = H$. For every y , $Hy \in \mathcal{C}(H)$, and for every $Hx \in \mathcal{C}(H)$,

$$(Hx)^\top (y - Hy) = x^\top (H - H^2)y = 0.$$

Thus Hy satisfies the defining conditions of the projection of y on $\mathcal{C}(H)$, proving the converse.

When $r = 0$, the empty matrix product is $QQ^\top = 0_{n \times n}$; the same identities hold with the empty-sum conventions of Unit 4. □

Example 6.2. Project $y = (2, 1, 0)^\top$ onto

$$W = \text{span}\{u = (1, -1, 1)^\top, v = (0, 1, 1)^\top\}.$$

Find the projection matrix, residual, and minimum squared distance as well.

- $u^\top v = 0$, $u^\top u = 3$, $v^\top v = 2$, and $u^\top y = v^\top y = 1$. Hence

$$p = \frac{1}{3}u + \frac{1}{2}v = \left(\frac{1}{3}, \frac{1}{6}, \frac{5}{6}\right)^\top, \quad e = y - p = \left(\frac{5}{3}, \frac{5}{6}, -\frac{5}{6}\right)^\top.$$

- The orthonormal basis is $u/\sqrt{3}, v/\sqrt{2}$, so

$$P = \frac{uu^\top}{3} + \frac{vv^\top}{2} = \frac{1}{6} \begin{pmatrix} 2 & -2 & 2 \\ -2 & 5 & 1 \\ 2 & 1 & 5 \end{pmatrix}.$$

Direct multiplication gives $Py = p$, $u^\top e = 0$, and $v^\top e = 0$.

- The squared residual norm is

$$\|e\|^2 = \frac{25}{9} + \frac{25}{36} + \frac{25}{36} = \frac{25}{6}.$$

Theorem 6.7 therefore gives, for every $w \in W$,

$$\|y - w\|^2 = \frac{25}{6} + \|p - w\|^2.$$

Thus p alone attains the minimum squared distance $25/6$.

Proposition 6.9. For any real $A : m \times p$,

$$\begin{aligned}\mathcal{C}(A)^\perp &= N(A^\top), & N(A)^\perp &= \mathcal{C}(A^\top), \\ N(A^\top A) &= N(A), & \text{rank}(A^\top A) &= \text{rank } A, \\ \mathcal{C}(A^\top A) &= \mathcal{C}(A^\top).\end{aligned}$$

The **Gram matrix** $A^\top A$ is invertible if and only if the columns of A are independent.

Proof. • For $A = [a_1 \cdots a_p]$, a column z is orthogonal to $\mathcal{C}(A)$ exactly when every $a_j^\top z = 0$, or $A^\top z = 0$. Applying this identity to $A^\top$ gives $\mathcal{C}(A^\top)^\perp = N(A)$; taking complements and using Theorem 6.5 gives the second identity.

- If $Ax = 0$, then $A^\top Ax = 0$. Conversely,

$$A^\top Ax = 0 \implies 0 = x^\top A^\top Ax = (Ax)^\top (Ax) = \|Ax\|^2 \implies Ax = 0.$$

The two matrices have the same domain $\mathbb{R}^p$ and the same kernel. Unit 3, Theorem 3.11, therefore gives equal ranks.

- Every $A^\top Ax$ is in $\mathcal{C}(A^\top)$, proving inclusion. The two column spaces have equal dimension, since $\text{rank}(A^\top A) = \text{rank } A = \text{rank } A^\top$ by Unit 3. The equal-dimension subspace criterion proves equality.
- By the square-matrix inverse criterion of Unit 2, $A^\top A$ is invertible exactly when its null space is $\{0\}$. The kernel identity makes this equivalent to $N(A) = \{0\}$, which is independence of the columns of A .

□

Proposition 6.10 (Consistency and the left null space). For $A : m \times p$ and $b \in \mathbb{R}^m$,

$$Ax = b \text{ is consistent} \iff z^\top b = 0 \text{ for every } z \in N(A^\top).$$

It suffices to test the vectors of any basis of $N(A^\top)$.

Proof. By Proposition 6.9 and the double-complement identity,

$$Ax = b \text{ is consistent} \iff b \in \mathcal{C}(A) \iff b \in N(A^\top)^\perp.$$

The last condition is exactly the stated orthogonality test. Testing a basis suffices because $(\sum_i c_i z_i)^\top b = \sum_i c_i z_i^\top b$. □

Example 6.3. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad W = \mathcal{C}(A).$$

Find bases for W and $W^\perp$. Compute the projection matrix P onto W both from an orthogonal basis of W and from $W^\perp$. For arbitrary $b \in \mathbb{R}^3$, find the residual $b - Pb$, the minimum of $\|b - Ax\|^2$, and the condition for $Ax = b$ to be consistent. In the consistent case, find its solution.

- Write

$$u = (1, 0, -1)^\top, \quad v = (0, 1, 2)^\top, \quad Ax = (x_1 + 2x_2)u + (3x_1 + 4x_2)v.$$

Conversely, prescribed coefficients λ_1, λ_2 arise from

$$x_1 = \lambda_2 - 2\lambda_1, \quad x_2 = (3\lambda_1 - \lambda_2)/2.$$

Thus $W = \text{span}(u, v)$. Their first two coordinates show that u, v are independent.

- To find $W^\perp = N(A^\top)$, solve

$$z_1 + 3z_2 + 5z_3 = 0, \quad 2z_1 + 4z_2 + 6z_3 = 0.$$

Subtract twice the first equation from the second: $z_2 = -2z_3$, and then $z_1 = z_3$. Therefore

$$W^\perp = \text{span}(n), \quad n = (1, -2, 1)^\top.$$

- Gram–Schmidt on u, v gives

$$u^\top u = 2, \quad u^\top v = -2, \quad w = v - \frac{u^\top v}{u^\top u} u = v + u = (1, 1, 1)^\top.$$

Since $u^\top w = 0$ and $w^\top w = 3$,

$$P = \frac{uu^\top}{2} + \frac{ww^\top}{3} = \frac{1}{6} \begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}.$$

- Alternatively, $n^\top n = 6$, so the complementary projection is

$$I_3 - P = \frac{nn^\top}{6} = \frac{1}{6} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

Subtraction from I_3 gives the same P .

- Put $\delta = n^\top b = b_1 - 2b_2 + b_3$. Then

$$e = b - Pb = \frac{\delta}{6}n, \quad \|e\|^2 = \frac{\delta^2}{36}n^\top n = \frac{\delta^2}{6}.$$

Every Ax lies in W , and Pb is attained by some Ax . The closest-point theorem therefore gives $\min_x \|b - Ax\|^2 = \delta^2/6$.

- The consistency criterion is $\delta = 0$. In that case, $b = b_1u + b_2v$, so

$$x_1 = b_2 - 2b_1, \quad x_2 = (3b_1 - b_2)/2.$$

These satisfy the first two equations, and $5x_1 + 6x_2 = -b_1 + 2b_2 = b_3$. The first two equations determine x_1, x_2 uniquely.

3. Least squares

Given $A : m \times p$ and $y \in \mathbb{R}^m$, the **least squares problem** asks for all $\beta \in \mathbb{R}^p$ minimizing

$$\|y - A\beta\|^2 = \sum_{i=1}^m \left(y_i - \sum_{j=1}^p a_{ij}\beta_j \right)^2.$$

The vectors $A\beta$ range over exactly $\mathcal{C}(A)$.

Theorem 6.11 (Normal equations). For every $A : m \times p$ and $y \in \mathbb{R}^m$, the equations

$$A^\top A\beta = A^\top y$$

are consistent. Their solutions are exactly the least squares minimizers. If β_0 is one solution, all solutions are

$$\beta = \beta_0 + z, \quad z \in N(A).$$

They all give the same fitted vector $\hat{y} = A\beta_0 = P_{\mathcal{C}(A)}y$.

- Proof.* • The projection $\hat{y} = P_{\mathcal{C}(A)}y$ exists and belongs to $\mathcal{C}(A)$, so $\hat{y} = A\beta_0$ for some β_0 . Its residual is orthogonal to $\mathcal{C}(A)$; by Proposition 6.9, $A^\top(y - A\beta_0) = 0$. This proves consistency.
- By Theorem 6.7, β minimizes the criterion exactly when $y - A\beta \perp \mathcal{C}(A)$, or $A^\top A\beta = A^\top y$. Projection uniqueness then gives $A\beta = \hat{y}$.
 - Subtracting the equations for β and β_0 gives $\beta - \beta_0 \in N(A^\top A) = N(A)$. Conversely, adding any $z \in N(A)$ preserves the fitted vector and the normal equations. This proves the full solution formula.

In particular, the coefficients are unique exactly when $N(A) = \{0\}$, even though the fitted vector is unique for every rank of A . $\square$

Proposition 6.12 (Full column rank). Let $A : m \times p$, $y \in \mathbb{R}^m$, and $\text{rank } A = p$. The unique least squares coefficient vector and the projection matrix onto $\mathcal{C}(A)$ are

$$\hat{\beta} = (A^\top A)^{-1} A^\top y, \quad P = A(A^\top A)^{-1} A^\top.$$

With $e = y - A\hat{\beta}$,

$$A^\top e = 0, \quad \|e\|^2 = y^\top (I_m - P)y, \quad \text{rank}(I_m - P) = m - p.$$

- Proof.* • The Gram matrix is invertible by Proposition 6.9. Solving the normal equations gives $\hat{\beta}$. Multiplication by A gives the displayed matrix for the unique fitted vector, hence for the projection.
- The normal equations give $A^\top e = 0$. Proposition 6.8 gives $P^\top = P = P^2$. Therefore

$$\begin{aligned} \|e\|^2 &= y^\top (I_m - P)^\top (I_m - P)y \\ &= y^\top (I_m - 2P + P^2)y = y^\top (I_m - P)y. \end{aligned}$$

- The image of $I_m - P$ is $\mathcal{C}(A)^\perp$, whose dimension is $m - \dim \mathcal{C}(A) = m - p$. $\square$

Example 6.4. Find the least squares coefficients, fitted vector, residual, projection matrix, and minimum sum of squares for

$$A = \begin{pmatrix} 0 & -2 \\ 3 & 2 \\ 6 & 4 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ -38/5 \\ 74/5 \end{pmatrix}.$$

Then append the column $a_3 = (-2, 1, 2)^\top$ to form $\tilde{A} = [A \ a_3]$, and determine all least squares coefficients for $\tilde{A}$. Compare the fitted vectors and minimum sums of squares.

- Direct multiplication gives

$$A^\top A = \begin{pmatrix} 45 & 30 \\ 30 & 24 \end{pmatrix}, \quad A^\top y = \begin{pmatrix} 66 \\ 38 \end{pmatrix}.$$

Since $\det(A^\top A) = 45 \cdot 24 - 30^2 = 180 \neq 0$, the Gram matrix is invertible and A has full column rank. The 2×2 inverse formula gives

$$(A^\top A)^{-1} = \begin{pmatrix} 2/15 & -1/6 \\ -1/6 & 1/4 \end{pmatrix}.$$

Thus

$$\hat{\beta} = \begin{pmatrix} 37/15 \\ -3/2 \end{pmatrix}, \quad \hat{y} = A\hat{\beta} = (3, 22/5, 44/5)^\top, \quad e = (0, -12, 6)^\top.$$

- Multiplying $A(A^\top A)^{-1}A^\top$ gives

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 2/5 \\ 0 & 2/5 & 4/5 \end{pmatrix}, \quad A^\top e = \begin{pmatrix} -36 + 36 \\ -24 + 24 \end{pmatrix} = 0, \quad \|e\|^2 = 180.$$

- If a_1, a_2 are the columns of A , then $a_3 = a_2 - a_1/3$. Hence $\mathcal{C}(\tilde{A}) = \mathcal{C}(A)$, so the fitted vector and minimum sum of squares remain $\hat{y}$ and 180. For $b = (b_1, b_2, b_3)^\top$,

$$\tilde{A}b = A \begin{pmatrix} b_1 - b_3/3 \\ b_2 + b_3 \end{pmatrix}.$$

Because A has independent columns, this equals $\hat{y}$ exactly when $b_1 - b_3/3 = 37/15$ and $b_2 + b_3 = -3/2$. Therefore all minimizing coefficients are

$$b = \begin{pmatrix} 37/15 + t/3 \\ -3/2 - t \\ t \end{pmatrix}, \quad t \in \mathbb{R}.$$

- The singular normal equations are explicitly

$$\begin{pmatrix} 45 & 30 & 15 \\ 30 & 24 & 14 \\ 15 & 14 & 9 \end{pmatrix} b = \begin{pmatrix} 66 \\ 38 \\ 16 \end{pmatrix}.$$

The third column of the coefficient matrix is its second minus one third of its first. The equations are nevertheless consistent: substitution of the displayed b gives $(66, 38, 16)^\top$ for every t . For instance $t = -1$ gives $b = (32/15, -1/2, -1)^\top$, with the same fit and residual.

Proposition 6.13 (Least squares by QR). Let $A = QR$ be the thin QR decomposition of a full-column-rank $m \times p$ matrix, $p \geq 1$, and put $P = QQ^\top$. For every $y \in \mathbb{R}^m$ and $\beta \in \mathbb{R}^p$,

$$\|y - A\beta\|^2 = \|(I_m - P)y\|^2 + \|Q^\top y - R\beta\|^2.$$

The unique least squares coefficients are obtained by back substitution in $R\hat{\beta} = Q^\top y$. The fitted vector and minimum sum of squares are

$$\hat{y} = QQ^\top y, \quad \|y - \hat{y}\|^2 = \|y\|^2 - \|Q^\top y\|^2.$$

Proof. • The columns of Q form an orthonormal basis of $\mathcal{C}(A)$, so P is its projection matrix. Decompose

$$y - QR\beta = (I_m - P)y + Q(Q^\top y - R\beta).$$

The two terms are orthogonal: the first is in $\mathcal{C}(Q)^\perp$ and the second in $\mathcal{C}(Q)$. Also $\|Qz\|^2 = z^\top Q^\top Qz = \|z\|^2$. Pythagoras now gives the displayed identity.

- Put $c = Q^\top y$. Since all $r_{jj} > 0$, back substitution determines exactly one vector:

$$\hat{\beta}_p = c_p/r_{pp}, \quad \hat{\beta}_j = \frac{c_j - \sum_{k=j+1}^p r_{jk}\hat{\beta}_k}{r_{jj}} \quad (j = p-1, \dots, 1).$$

This makes $\|c - R\beta\|^2$ zero; every other β makes it positive. Thus it gives the unique minimum, with $A\hat{\beta} = Qc = Py$.

- Applying Pythagoras to $y = Py + (I_m - P)y$ gives

$$\|y\|^2 = \|Py\|^2 + \|(I_m - P)y\|^2, \quad \|Py\|^2 = \|Q^\top y\|^2.$$

Rearrangement gives the stated minimum sum of squares. □

Example 6.5. Find the thin QR decomposition of A , then use it to find the least squares coefficients, fitted vector, residual, and minimum sum of squares:

$$A = \begin{pmatrix} 6 & -2 \\ 0 & 4 \\ -2 & 4 \\ 3 & 2 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

- Let a_1, a_2 be the columns. Then

$$\|a_1\|^2 = 49, \quad a_1^\top a_2 = -14, \quad q_1 = \frac{1}{7}(6, 0, -2, 3)^\top, \quad r_{12} = q_1^\top a_2 = -2.$$

The second residual is

$$u_2 = a_2 + 2q_1 = \frac{1}{7}(-2, 28, 24, 20)^\top, \quad \|u_2\|^2 = \frac{4 + 784 + 576 + 400}{49} = 36.$$

Directly, $a_1^\top u_2 = (-12 - 48 + 60)/7 = 0$. Both residuals are nonzero, and $a_1 = 7q_1$, $a_2 = -2q_1 + 6q_2$, where $q_2 = u_2/6$. Hence A has independent columns and its QR factors are

$$Q = \begin{pmatrix} 6/7 & -1/21 \\ 0 & 2/3 \\ -2/7 & 4/7 \\ 3/7 & 10/21 \end{pmatrix}, \quad R = \begin{pmatrix} 7 & -2 \\ 0 & 6 \end{pmatrix}.$$

- Since y is the second coordinate vector,

$$Q^\top y = \begin{pmatrix} 0 \\ 2/3 \end{pmatrix}, \quad 6\hat{\beta}_2 = 2/3, \quad 7\hat{\beta}_1 - 2\hat{\beta}_2 = 0.$$

Back substitution gives $\hat{\beta}_2 = 1/9$, $\hat{\beta}_1 = 2/63$. Thus

$$\hat{y} = Q \begin{pmatrix} 0 \\ 2/3 \end{pmatrix} = \frac{1}{63} \begin{pmatrix} -2 \\ 28 \\ 24 \\ 20 \end{pmatrix}, \quad e = y - \hat{y} = \frac{1}{63} \begin{pmatrix} 2 \\ 35 \\ -24 \\ -20 \end{pmatrix}.$$

- The residual is orthogonal to the columns of A :

$$A^\top e = \frac{1}{63} \begin{pmatrix} 12 + 48 - 60 \\ -4 + 140 - 96 - 40 \end{pmatrix} = 0.$$

The QR identity gives

$$\min_{\beta} \|y - A\beta\|^2 = 1 - (2/3)^2 = 5/9.$$

Directly, $\|e\|^2 = (4 + 1225 + 576 + 400)/63^2 = 5/9$.

Exercises and solutions

Suggested practice: Exercises 6.1, 6.11, 6.14, 6.13, 6.20, and 6.26. All vectors and matrices in these exercises are real.

Dot products and norms

Exercise 6.1 (Computing dot products). Let

$$x = (1, 2, -3)^\top, \quad y = (4, -5, 1)^\top, \quad z = (1, 1, 1)^\top, \quad w = (3, \alpha, 1)^\top.$$

Compute $x^\top y, x^\top z, y^\top z$, and find all α such that $y^\top w = 0$.

Solution. Directly,

$$x^\top y = 4 - 10 - 3 = -9, \quad x^\top z = 1 + 2 - 3 = 0, \quad y^\top z = 4 - 5 + 1 = 0.$$

Since $y^\top w = 12 - 5\alpha + 1 = 13 - 5\alpha$, the last condition holds exactly when $\alpha = 13/5$.

Exercise 6.2 (Normalization). For $x \neq 0$, put $x_\circ = x/\|x\|$.

- Prove that $\|x_\circ\| = 1$.
- Find the norms of $a = (1, 2)^\top, b = (1, 0)^\top, c = (3, 4)^\top$.
- Normalize a, b, c .
- Prove that x and λx have the same normalized vector for every $\lambda > 0$.

Solution.

- Positivity gives $\|x\| > 0$, and homogeneity gives $\|x/\|x\|\| = \|x\|/\|x\| = 1$.
- The squared norms are $1 + 4 = 5, 1, 9 + 16 = 25$, respectively. Thus the norms are $\sqrt{5}, 1, 5$.
- The answers are

$$a_\circ = \frac{1}{\sqrt{5}}(1, 2)^\top, \quad b_\circ = (1, 0)^\top, \quad c_\circ = \frac{1}{5}(3, 4)^\top.$$

- Since $\|\lambda x\| = |\lambda|\|x\|$,

$$(\lambda x)_\circ = \frac{\lambda}{|\lambda|} x_\circ = x_\circ \quad (\lambda > 0).$$

For $\lambda < 0$ this formula gives $-x_\circ$; for $\lambda = 0$ normalization is undefined.

Exercise 6.3 (All orthogonal columns). Find all columns of $\mathbb{R}^2$ orthogonal to (a) $a = (1, 2)^\top$, and (b) $b = (1, 0)^\top$.

Solution. Writing $x = (x_1, x_2)^\top$, condition (a) is $x_1 + 2x_2 = 0$. All its solutions are $x = \lambda(2, -1)^\top$, $\lambda \in \mathbb{R}$. Condition (b) is $x_1 = 0$, whose solutions are $x = \lambda(0, 1)^\top$. Substitution verifies each displayed family; the scalar equations show that no other columns occur.

Exercise 6.4 (Failure of transitivity). Does $x \perp y$ and $y \perp z$ imply $x \perp z$? Give a counterexample if the implication fails.

Solution. Take the three columns, in this order,

$$x = (1, 2, -3)^\top, \quad y = (1, 1, 1)^\top, \quad z = (4, -5, 1)^\top.$$

Then $x^\top y = 0, y^\top z = 0$, but $x^\top z = -9 \neq 0$. More generally, if $x \neq 0$ and $x \perp y$, taking $z = x$ gives the first two orthogonality conditions but $x^\top z = \|x\|^2 > 0$.

Exercise 6.5 (The parallelogram identity *). An inner product on $\mathbb{R}^m$ is a function $h : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}$ that is symmetric, linear in each variable, and satisfies $h(x, x) > 0$ for $x \neq 0$. Linearity gives $h(0, 0) = h(0 + 0, 0) = 2h(0, 0)$, hence $h(0, 0) = 0$. Write $\|x\|_h = \sqrt{h(x, x)}$. Prove, using these properties, that

$$\|x + y\|_h^2 + \|x - y\|_h^2 = 2\|x\|_h^2 + 2\|y\|_h^2.$$

Solution. Bilinearity and symmetry give

$$h(x + y, x + y) = h(x, x) + 2h(x, y) + h(y, y),$$

$$h(x - y, x - y) = h(x, x) - 2h(x, y) + h(y, y).$$

Adding cancels the two cross terms and proves the identity. The calculation applies to every such h , including $h(x, y) = x^\top y$.

Exercise 6.6 (Recovering an inner product). For an inner product h as in Exercise 6.5, prove

$$h(x, y) = \frac{1}{4}(\|x + y\|_h^2 - \|x - y\|_h^2).$$

Solution. Subtracting the two expansions in Exercise 6.5 gives

$$\|x + y\|_h^2 - \|x - y\|_h^2 = 4h(x, y).$$

Thus two inner products inducing the same norm have the same values on every pair x, y .

Exercise 6.7 (A norm not induced by an inner product *). Let $m \geq 2$ and put $\nu(x) = \sum_{i=1}^m |x_i|$.

(a) Prove the three norm properties:

$$\nu(cx) = |c|\nu(x), \quad \nu(x) \geq 0 \text{ with equality iff } x = 0, \quad \nu(x + y) \leq \nu(x) + \nu(y).$$

(b) Prove that no inner product h on $\mathbb{R}^m$ satisfies $\nu(x) = \sqrt{h(x, x)}$ for every x .

Solution.

(a) Each term $|x_i|$ is nonnegative; their sum is zero exactly when every $x_i = 0$. Also $|cx_i| = |c||x_i|$, which proves homogeneity. For real s, t , the inequalities $-|s| \leq s \leq |s|$ and $-|t| \leq t \leq |t|$ give

$$-(|s| + |t|) \leq s + t \leq |s| + |t|, \quad \text{hence } |s + t| \leq |s| + |t|.$$

Apply this to each coordinate and add.

(b) Take

$$x = (1, 1, 0, \dots, 0)^\top, \quad y = (1, -1, 0, \dots, 0)^\top.$$

Then $\nu(x) = \nu(y) = \nu(x + y) = \nu(x - y) = 2$, so

$$\nu(x + y)^2 + \nu(x - y)^2 = 8 \neq 16 = 2\nu(x)^2 + 2\nu(y)^2.$$

This contradicts the necessary identity proved in Exercise 6.5. The restriction $m \geq 2$ matters: in $\mathbb{R}$, $\nu(x) = |x|$ is induced by $h(x, y) = xy$.

Exercise 6.8 (The Frobenius norm $*$). For $A, B \in \mathbb{R}^{m \times n}$, define

$$\langle A, B \rangle_F = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij} = \text{tr}(A^\top B), \quad \|A\|_F = \left(\sum_{i,j} a_{ij}^2 \right)^{1/2}.$$

Prove (a) $\|cA\|_F = |c|\|A\|_F$; (b) $\|A\|_F \geq 0$, with equality iff $A = 0$; (c) $\|A + B\|_F \leq \|A\|_F + \|B\|_F$.

Solution. Let $a, b \in \mathbb{R}^{mn}$ list the entries of A, B in the same order, column by column. Directly from the definitions,

$$\langle A, B \rangle_F = a^\top b, \quad \|A\|_F = \|a\|, \quad \|B\|_F = \|b\|, \quad \|A + B\|_F = \|a + b\|.$$

The trace equality follows because $(A^\top B)_{jj} = \sum_i a_{ij} b_{ij}$. The list for cA is ca , and $a = 0$ iff $A = 0$. The three assertions therefore follow from the proved vector norm properties. The same correspondence proves symmetry, bilinearity, and strict positivity of $\langle \cdot, \cdot \rangle_F$; these properties will also be used for matrix subspaces below.

Exercise 6.9 (Zero and an orthogonal complement). (a) Prove that the zero column is orthogonal to every column.

(b) Find all columns in $\mathbb{R}^3$ orthogonal to $(2, 2, -1)^\top$.

Solution.

(a) $0^\top y = \sum_i 0y_i = 0$ for every y .

(b) Writing $y = (y_1, y_2, y_3)^\top$, the necessary and sufficient equation is $2y_1 + 2y_2 - y_3 = 0$. Thus all solutions, with unrestricted $s, t \in \mathbb{R}$, are

$$y = (s, t, 2s + 2t)^\top = s(1, 0, 2)^\top + t(0, 1, 2)^\top.$$

Both generators have dot product zero with the given column, verifying the reverse inclusion.

Orthonormal bases

Exercise 6.10 (Two orthogonality conditions). What is the dimension of the subspace of $\mathbb{R}^6$ orthogonal to both

$$a = (1, 1, -2, 3, 4, 5)^\top, \quad b = (0, 0, 1, 1, 0, 7)^\top?$$

Solution. In a relation $sa + tb = 0$, coordinate 1 gives $s = 0$, and coordinate 3 then gives $t = 0$. Hence $\dim \text{span}(a, b) = 2$. Being orthogonal to a, b is equivalent, by linearity, to being orthogonal to every $sa + tb$. The required space is $\text{span}(a, b)^\perp$, of dimension $6 - 2 = 4$.

Exercise 6.11 (Two bases in $\mathbb{R}^3$). Find an orthonormal basis for the span of each list:

(a) $(1, 1, -1)^\top, (1, 0, 1)^\top$.

(b) $(2, 1, 1)^\top, (1, 3, -1)^\top$.

Solution.

(a) Put $a = (1, 1, -1)^\top, b = (1, 0, 1)^\top$. Then $a^\top b = 1 - 1 = 0$, $\|a\|^2 = 3$, and $\|b\|^2 = 2$. The two nonzero orthogonal columns are independent, so

$$\frac{a}{\sqrt{3}}, \quad \frac{b}{\sqrt{2}}$$

is an orthonormal list with the required span.

(b) Put $a = (2, 1, 1)^\top$, $b = (1, 3, -1)^\top$. Here $a^\top a = 6$ and $a^\top b = 4$, so

$$u = b - \frac{4}{6}a = \frac{1}{3}(-1, 7, -5)^\top, \quad a^\top(-1, 7, -5)^\top = -2 + 7 - 5 = 0.$$

The squared norm of $(-1, 7, -5)^\top$ is $1 + 49 + 25 = 75$. An answer is

$$\frac{1}{\sqrt{6}}(2, 1, 1)^\top, \quad \frac{1}{\sqrt{75}}(-1, 7, -5)^\top.$$

Both are nonzero and orthogonal. Since $u = b - \frac{2}{3}a$ and $b = u + \frac{2}{3}a$, the new and original lists have the same span.

Exercise 6.12 (Two bases in $\mathbb{R}^4$). Find an orthonormal basis for the span of each list:

(a) $(1, 2, 1, 0)^\top$, $(1, 2, 3, 1)^\top$.

(b) $(1, 1, 0, 0)^\top$, $(1, -1, 1, 1)^\top$, $(-1, 0, 2, 1)^\top$.

Solution.

(a) Call the two given columns a, b , in order. Then $a^\top a = 6$, $a^\top b = 8$, so

$$u = b - \frac{8}{6}a = \frac{1}{3}(-1, -2, 5, 3)^\top.$$

The numerator has squared norm $1 + 4 + 25 + 9 = 39$, and its dot product with a is $-1 - 4 + 5 = 0$. The answer is

$$\frac{1}{\sqrt{6}}(1, 2, 1, 0)^\top, \quad \frac{1}{\sqrt{39}}(-1, -2, 5, 3)^\top.$$

The relations $u = b - \frac{4}{3}a$, $b = u + \frac{4}{3}a$ prove span equality.

(b) Write the original columns as a, b, c . Their relevant products are

$$a^\top b = 0, \quad a^\top a = 2, \quad b^\top b = 4, \quad c^\top a = -1, \quad c^\top b = 2.$$

Hence

$$u = c + \frac{1}{2}a - \frac{1}{2}b = \frac{1}{2}(-2, 2, 3, 1)^\top.$$

For $d = (-2, 2, 3, 1)^\top$, $a^\top d = 0$, $b^\top d = -2 - 2 + 3 + 1 = 0$, and $d^\top d = 18$. An answer is

$$\frac{1}{\sqrt{2}}(1, 1, 0, 0)^\top, \quad \frac{1}{2}(1, -1, 1, 1)^\top, \quad \frac{1}{\sqrt{18}}(-2, 2, 3, 1)^\top.$$

The formulas for u and $c = u - \frac{1}{2}a + \frac{1}{2}b$ prove equality of the old and new spans. Nonzero orthogonal columns are independent, so both displayed lists are bases.

Exercise 6.13 (Gram–Schmidt with dependent inputs). Let $A_1, \dots, A_k \in \mathbb{R}^{m \times n}$, $k \geq 1$, possibly dependent or zero. Use the inner product $\langle A, B \rangle_F = \text{tr}(A^\top B)$.

(a) Prove all three assertions:

(1) Scalars c_{ij} , $i < j$, can be chosen so that

$$B_1 = A_1, \quad B_j = A_j - \sum_{i < j} c_{ij} B_i \quad (2 \leq j \leq k)$$

are pairwise orthogonal.

(2) For every nonzero B_i , orthogonality forces the unique value

$$c_{ij} = \frac{\langle A_j, B_i \rangle_F}{\langle B_i, B_i \rangle_F} \quad (i < j).$$

(3) The number of nonzero B_i 's equals $\dim \text{span}(A_1, \dots, A_k)$.

(b) Describe a procedure yielding an orthonormal basis of $\text{span}(A_1, \dots, A_k)$.

Solution.

(a) Suppose $B_1, \dots, B_{j-1}$ have been chosen and are pairwise orthogonal; the assertion is empty for $j = 1$. For $i < j$, orthogonality of the preceding matrices gives

$$\langle B_j, B_i \rangle_F = \langle A_j, B_i \rangle_F - c_{ij} \|B_i\|_F^2.$$

If $B_i \neq 0$, its squared norm is positive, so this vanishes for exactly the stated coefficient. If $B_i = 0$, the equation is $0 = 0$; any coefficient works, and we choose $c_{ij} = 0$. Induction proves (1) and (2), without division by zero.

For (3), induction on j using the defining expression for B_j shows that each $B_j \in \text{span}(A_1, \dots, A_j)$. Conversely

$$A_j = B_j + \sum_{i < j} c_{ij} B_i$$

puts each A_j in $\text{span}(B_1, \dots, B_j)$. Thus every prefix has the same span. If a linear combination of the nonzero B_i 's is zero, taking its inner product with $B_\ell \neq 0$ gives its coefficient times $\|B_\ell\|_F^2 = 0$; each coefficient is therefore zero. The nonzero outputs form a basis, proving the dimension assertion.

(b) Compute these residuals, discard exactly the zero outputs, and divide every remaining B_i by $\|B_i\|_F$. This preserves their span and makes the basis orthonormal. If every input is zero, every output is zero and the resulting empty list is the basis of the zero space.

Orthogonal matrices and ranks

Exercise 6.14 (Rectangular orthonormality). (a) Let $A \in \mathbb{R}^{m \times n}$, with $m < n$. Prove that no $B \in \mathbb{R}^{m \times n}$ can satisfy $B^\top A = I_n$.

(b) Give an $m \times n$ example, $m \neq n$, satisfying $AA^\top = I_m$ or $A^\top A = I_n$, but not both.

Solution.

(a) If $B^\top A = I_n$, the product-rank bound would give

$$n = \text{rank}(B^\top A) \leq \text{rank } A \leq m < n.$$

This is impossible.

(b) Take $n > 1$, let e_1 be the first standard basis column of $\mathbb{R}^n$, and set $A = e_1^\top$, so $m = 1$. Then $AA^\top = e_1^\top e_1 = I_1$. However, $A^\top A = e_1 e_1^\top \neq I_n$, since its $(2, 2)$ entry is zero. Transposing this example gives the other alternative.

Exercise 6.15 (Orthogonal matrices). Let $A \in \mathbb{R}^{n \times n}$ satisfy $A^\top A = I_n$. Prove (a) A is invertible; (b) $A^{-1} = A^\top$; (c) $A^\top$ is orthogonal; (d) AB is orthogonal whenever $B \in \mathbb{R}^{n \times n}$ is orthogonal.

Solution. For (a) and (b), the square one-sided inverse criterion of Unit 2 applied to $A^\top A = I_n$ gives $A^{-1} = A^\top$ and $AA^\top = I_n$. For (c), writing $C = A^\top$ gives $C^\top C = AA^\top = I_n$. For (d),

$$(AB)^\top(AB) = B^\top A^\top AB = B^\top B = I_n.$$

For completeness, the reverse product is also $(AB)(AB)^\top = ABB^\top A^\top = AA^\top = I_n$.

Exercise 6.16 (Preserving one norm or every norm). (a) Prove that an orthogonal $Q \in \mathbb{R}^{n \times n}$ satisfies $\|Qx\| = \|x\|$ for every $x \in \mathbb{R}^n$.

(b) For a particular nonzero x , does $\|Qx\| = \|x\|$ imply that Q is orthogonal?

(c) Suppose a fixed square Q , independent of x , satisfies $\|Qx\| = \|x\|$ for every $x \in \mathbb{R}^n$. Prove that Q is orthogonal.

Solution.

(a) $\|Qx\|^2 = x^\top Q^\top Qx = x^\top x = \|x\|^2$; both norms are nonnegative, so they are equal.

(b) For $n > 1, x \neq 0$, take

$$Q = \frac{xx^\top}{x^\top x}.$$

Then $Qx = x$, $Q^\top = Q$, and $Q^2 = Q$, by direct multiplication. Its columns are multiples of x , and $Qx = x \neq 0$; hence $\text{rank } Q = 1 < n$. Thus $Q^\top Q = Q \neq I_n$, so Q is not orthogonal. For $n = 1$, however, the answer is yes: if $x \neq 0$ and $|qx| = |x|$, then $|q| = 1$, hence $q^2 = 1$.

(c) Put $H = I_n - Q^\top Q$, a fixed symmetric matrix. The hypothesis gives $x^\top Hx = 0$ for every x . Taking $x = e_i$ gives $h_{ii} = 0$. For $i \neq j$, taking $x = e_i + e_j$ then gives

$$0 = h_{ii} + h_{ij} + h_{ji} + h_{jj} = 2h_{ij}.$$

Every entry of H is zero, so $Q^\top Q = I_n$.

Exercise 6.17 (Rank equalities *). For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, prove

$$\text{rank}(A^\top AB) = \text{rank}(AB) = \text{rank}(ABB^\top).$$

Solution. Proposition 6.9 gives $\text{rank}(X^\top X) = \text{rank } X$. Applying it to $X^\top$, and using the equality of row and column ranks, also gives $\text{rank}(XX^\top) = \text{rank}(X^\top) = \text{rank } X$. The product-rank bounds now yield

$$\text{rank}(AB) = \text{rank}(B^\top A^\top AB) \leq \text{rank}(A^\top AB) \leq \text{rank}(AB),$$

and

$$\text{rank}(AB) = \text{rank}(ABB^\top A^\top) \leq \text{rank}(ABB^\top) \leq \text{rank}(AB).$$

Each middle rank is trapped between two equal quantities. No full-rank assumption on A or B is required.

Identifying projection matrices

Exercise 6.18 (A sum of idempotents). Let $A, B \in \mathbb{R}^{n \times n}$ satisfy $A^2 = A, B^2 = B$.

- (a) Prove that $A + B$ is idempotent if and only if $AB = BA = 0$.
- (b) If $A + B$ is idempotent, prove $\text{rank}(A + B) = \text{rank } A + \text{rank } B$.

Solution.

- (a) If $AB = BA = 0$, expanding gives $(A + B)^2 = A + B$. Conversely, idempotence of the sum gives $AB + BA = 0$. Multiplying this equation on the left and on the right by A gives

$$AB + ABA = 0, \quad ABA + BA = 0.$$

Subtracting gives $AB = BA$. Thus $2AB = 0$, and over $\mathbb{R}$ we obtain $AB = BA = 0$.

- (b) Put $U = \mathcal{C}(A)$, $V = \mathcal{C}(B)$. Certainly $\mathcal{C}(A + B) \subseteq U + V$. For every $u = Ax$, $v = By$, part (a) gives

$$(A + B)(u + v) = A^2x + BAx + AB y + B^2y = u + v.$$

Hence $U + V \subseteq \mathcal{C}(A + B)$. If $z = Ax = By \in U \cap V$, then $Az = z$, but also $Az = AB y = 0$; therefore $z = 0$. The sum is direct, so the dimension formula gives

$$\text{rank}(A + B) = \dim(U \oplus V) = \dim U + \dim V.$$

Exercise 6.19 (Two complementary column blocks). Let $A = [A_1 \ A_2]$ be $n \times n$. Suppose

$$\text{rank } A_1 + \text{rank } A_2 = n, \quad A_1^\top A_2 = 0.$$

Prove that the two Gram inverses in the following expression exist, and that

$$A_1(A_1^\top A_1)^{-1}A_1^\top + A_2(A_2^\top A_2)^{-1}A_2^\top = I_n.$$

If a block has no columns, interpret its entire term as the zero matrix.

Solution. If $n = 0$, both terms are the empty zero matrix by the stated convention, so the identity holds. Now let $n > 0$. If one block is empty, the other is a square full-rank matrix C , hence invertible. The inverse product rule gives

$$C(C^\top C)^{-1}C^\top = CC^{-1}(C^\top)^{-1}C^\top = I_n.$$

Together with the zero term from the empty block, this proves the formula.

Now suppose both blocks are nonempty, with n_i columns in A_i . Since $n_1 + n_2 = n$ and $\text{rank } A_i \leq n_i$, the rank-sum hypothesis forces $\text{rank } A_i = n_i$ for both i . Thus $G_i = A_i^\top A_i$ is invertible by Proposition 6.9. If $A_1 x + A_2 y = 0$, multiplication by $A_1^\top$ gives $G_1 x = 0$, so $x = 0$; full column rank of A_2 then gives $y = 0$. Hence A is invertible.

Orthogonality and transposition give

$$A^\top A = \begin{pmatrix} G_1 & 0 \\ 0 & G_2 \end{pmatrix}, \quad (A^\top A)^{-1} = \begin{pmatrix} G_1^{-1} & 0 \\ 0 & G_2^{-1} \end{pmatrix};$$

the inverse formula is checked by multiplication in both orders. The inverse product rule gives $(A^\top A)^{-1} = A^{-1}(A^\top)^{-1}$. Therefore

$$I_n = A(A^\top A)^{-1}A^\top = A_1 G_1^{-1} A_1^\top + A_2 G_2^{-1} A_2^\top.$$

Exercise 6.20 (Two projection criteria). Let $A \in \mathbb{R}^{n \times n}$ be symmetric and idempotent, and let $B \in \mathbb{R}^{n \times m}$ have rank m .

(a) If $AB = B$ and $\text{rank } A = \text{rank } B$, prove

$$A = B(B^\top B)^{-1}B^\top.$$

(b) If $AB = 0$ and $\text{rank } A + \text{rank } B = n$, prove

$$A = I_n - B(B^\top B)^{-1}B^\top.$$

Solution. The Gram inverse exists by Proposition 6.9.

(a) The equation $AB = B$ shows that $\mathcal{C}(B) \subseteq \mathcal{C}(A)$. Their dimensions are equal, so these spaces coincide. Proposition 6.8 says A projects orthogonally onto $\mathcal{C}(A)$; Proposition 6.12 identifies $B(B^\top B)^{-1}B^\top$ as the projection onto $\mathcal{C}(B)$. Uniqueness of the projection of every vector gives equality of the matrices.

(b) Put $C = I_n - A$. Directly, $C^\top = C$ and $C^2 = C$, while $CB = B$. By Proposition 6.8, $\mathcal{C}(C) = \mathcal{C}(A)^\perp$; the complement dimension formula gives

$$\text{rank } C = n - \text{rank } A = \text{rank } B.$$

Part (a), applied to C, B , gives $C = B(B^\top B)^{-1}B^\top$, which is the required formula.

Inequalities and equality conditions

Exercise 6.21 (Two matrix proofs of Schwarz). For $a, b \in \mathbb{R}^n$, prove

$$(a^\top b)^2 \leq (a^\top a)(b^\top b),$$

with equality exactly when a, b are linearly dependent. Give a proof using each of the following matrices:

(a) $H = ab^\top - ba^\top$.

(b) $P = I_n - bb^\top / (b^\top b)$, for $b \neq 0$, treating $b = 0$ separately.

Solution.

(a) Expanding each entry gives

$$\begin{aligned} \|H\|_F^2 &= \sum_{i=1}^n \sum_{j=1}^n (a_i b_j - b_i a_j)^2 \\ &= \left(\sum_i a_i^2 \right) \left(\sum_j b_j^2 \right) + \left(\sum_i b_i^2 \right) \left(\sum_j a_j^2 \right) - 2 \left(\sum_i a_i b_i \right) \left(\sum_j a_j b_j \right) \\ &= 2(a^\top a)(b^\top b) - 2(a^\top b)^2 \geq 0. \end{aligned}$$

Equality holds exactly when $a_i b_j = b_i a_j$ for every i, j . If $b \neq 0$, choose j with $b_j \neq 0$; these equations give $a_i = (a_j / b_j) b_i$ for every i , so a is a multiple of b . If $b = 0$, equality holds for every a . Conversely, either $b = 0$ or $a = cb$ gives $H = 0$. These are exactly the dependent pairs.

(b) If $b = 0$, both sides are zero. For $b \neq 0$, put $E = bb^\top / (b^\top b)$. Then $E^\top = E$ and

$$E^2 = \frac{b(b^\top b)b^\top}{(b^\top b)^2} = E.$$

Consequently $P^\top = P$, $P^2 = P$, and

$$(a^\top a)(b^\top b) - (a^\top b)^2 = (b^\top b)a^\top P a = (b^\top b)\|Pa\|^2 \geq 0.$$

Equality is equivalent to $Pa = 0$, or $a = ((b^\top a) / (b^\top b))b$. Conversely, this relation makes $Pa = 0$.

Exercise 6.22 (Three trace inequalities). For $A, B \in \mathbb{R}^{m \times n}$, prove all three inequalities and their equality conditions:

(a)

$$\left(\operatorname{tr}(A^\top B)\right)^2 \leq \operatorname{tr}(A^\top A) \operatorname{tr}(B^\top B),$$

with equality iff one of A, B is a scalar multiple of the other.

(b)

$$\operatorname{tr}\left((A^\top B)^2\right) \leq \operatorname{tr}(A^\top A B^\top B),$$

with equality iff $AB^\top$ is symmetric.

(c)

$$\operatorname{tr}\left((A^\top B)^2\right) \leq \operatorname{tr}(AA^\top BB^\top),$$

with equality iff $A^\top B$ is symmetric.

Solution.

(a) List the entries of A, B in the same order as $a, b \in \mathbb{R}^{mn}$. Then the three traces are $a^\top b, a^\top a, b^\top b$. Theorem 6.1 gives the inequality, with equality iff $b = 0$ or $a = cb$ for some $c \in \mathbb{R}$. In matrix terms this is $B = 0$ or $A = cB$, exactly the stated condition that one is a multiple of the other.

(b) For any real square C , the trace laws give

$$\begin{aligned} \|C - C^\top\|_F^2 &= \operatorname{tr}(C^\top C) - \operatorname{tr}((C^\top)^2) - \operatorname{tr}(C^2) + \operatorname{tr}(CC^\top) \\ &= 2[\operatorname{tr}(C^\top C) - \operatorname{tr}(C^2)]. \end{aligned}$$

Thus $\operatorname{tr}(C^2) \leq \operatorname{tr}(C^\top C)$, with equality iff $C = C^\top$. Take $C = AB^\top$. By cyclicity and transposition,

$$\begin{aligned} \operatorname{tr}(C^2) &= \operatorname{tr}(B^\top AB^\top A) = \operatorname{tr}\left((A^\top B)^2\right), \\ \operatorname{tr}(C^\top C) &= \operatorname{tr}(BA^\top AB^\top) = \operatorname{tr}(A^\top AB^\top B). \end{aligned}$$

The preceding identity proves (b), including its exact equality condition.

(c) Apply the same identity to $C = A^\top B$. Now

$$\operatorname{tr}(C^\top C) = \operatorname{tr}(B^\top AA^\top B) = \operatorname{tr}(AA^\top BB^\top).$$

Equality holds exactly when this C is symmetric, proving (c).

Exercise 6.23 (A bound for a sum). For $x_1, \dots, x_n \in \mathbb{R}$, $n \geq 1$, prove

$$\left(\sum_{i=1}^n x_i\right)^2 \leq n \sum_{i=1}^n x_i^2,$$

with equality iff $x_1 = \dots = x_n$.

Solution. Let $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^n$ and $M = I_n - \mathbf{1}\mathbf{1}^\top/n$. Since $\mathbf{1}^\top \mathbf{1} = n$, multiplication gives

$$M^\top = M, \quad M^2 = I_n - \frac{2}{n}\mathbf{1}\mathbf{1}^\top + \frac{1}{n}\mathbf{1}\mathbf{1}^\top = M.$$

Therefore

$$0 \leq \|Mx\|^2 = x^\top Mx = \sum_i x_i^2 - \frac{1}{n} \left(\sum_i x_i\right)^2.$$

Multiplying by $n > 0$ proves the bound. Equality holds iff $Mx = 0$, or $x_i = (\sum_j x_j)/n$ for every i , which is equivalent to all the entries being equal.

Matrix subspaces and least squares

Exercise 6.24 (A dimension obstruction). Let U, V be subspaces of $\mathbb{R}^{m \times n}$, with inner product $\langle A, B \rangle_F = \text{tr}(A^\top B)$. Prove that if $\dim V > \dim U$, then V contains a nonzero matrix orthogonal to every member of U .

Solution. Take bases $A_1, \dots, A_r$ of U and $B_1, \dots, B_s$ of V , where $s > r$. If $r = 0$, any $B_1 \neq 0$ has the required property, since $U = \{0\}$. If $r > 0$, define $H \in \mathbb{R}^{r \times s}$ by

$$h_{ij} = \langle A_i, B_j \rangle_F.$$

There are more unknowns than equations in $Hx = 0$, so a nonzero solution $x = (x_1, \dots, x_s)^\top$ exists. Put $C = \sum_{j=1}^s x_j B_j \in V$. Independence of the B_j 's gives $C \neq 0$, and

$$\langle A_i, C \rangle_F = \sum_j h_{ij} x_j = (Hx)_i = 0 \quad (1 \leq i \leq r).$$

Every $A \in U$ is a linear combination of the A_i 's; bilinearity then gives $\langle A, C \rangle_F = 0$, as required.

Exercise 6.25 (Projecting a matrix column by column). Let $U \subseteq \mathbb{R}^m$ be a subspace and define

$$\mathcal{M} = \{[w_1 \ \cdots \ w_n] : w_j \in U \text{ for every } j\} \subseteq \mathbb{R}^{m \times n}.$$

Let $Y \in \mathbb{R}^{m \times n}$, and let $X \in \mathbb{R}^{m \times p}$ have columns spanning U , without assuming they are independent. Use $\langle A, B \rangle_F = \text{tr}(A^\top B)$. Here the orthogonal projection is the matrix $Z \in \mathcal{M}$ with $\langle Y - Z, W \rangle_F = 0$ for every $W \in \mathcal{M}$. Show that this projection exists, is unique, and satisfies

$$Z = XB_*, \quad \text{for every solution } B_* \in \mathbb{R}^{p \times n} \text{ of } X^\top X B = X^\top Y.$$

Solution. The zero matrix is in $\mathcal{M}$, and linear combinations of its members stay in $\mathcal{M}$, because this holds for each column in U . Thus $\mathcal{M}$ is a subspace. Write $Y = [y_1 \ \cdots \ y_n]$, put $v_j = P_U y_j$, and let $V = [v_1 \ \cdots \ v_n] \in \mathcal{M}$. For every $W = [w_1 \ \cdots \ w_n] \in \mathcal{M}$,

$$\langle Y - V, W \rangle_F = \sum_{j=1}^n (y_j - v_j)^\top w_j = 0.$$

Consequently, for any $W \in \mathcal{M}$, the two terms in $Y - W = (Y - V) + (V - W)$ are orthogonal, and expansion gives

$$\|Y - W\|_F^2 = \|Y - V\|_F^2 + \|V - W\|_F^2.$$

It follows that V is the unique closest matrix. If another $Z' \in \mathcal{M}$ has residual orthogonal to $\mathcal{M}$, put $D = Z' - V \in \mathcal{M}$. Then

$$\|D\|_F^2 = \langle Y - V, D \rangle_F - \langle Y - Z', D \rangle_F = 0,$$

so $D = 0$ and $Z' = V$. This also proves uniqueness under the defining projection conditions; thus $Z = V$.

By Theorem 6.11, each system $X^\top X b_j = X^\top y_j$ is consistent, even if X has dependent columns. Stacking one solution of each system gives a solution B . For every solution B_* , the same theorem gives $X b_{*j} = P_U y_j = v_j$ for all j . Therefore $X B_* = V = Z$, independently of which coefficient solution is used.

Exercise 6.26 (Recognizing an orthonormal basis). Let $V \subseteq \mathbb{R}^n$ be a subspace and let $v_1, \dots, v_r \in V$ be an orthonormal list. Suppose that

$$\|v\|^2 = \sum_{i=1}^r (v^\top v_i)^2 \quad \text{for every } v \in V.$$

Prove that $v_1, \dots, v_r$ is a basis of V .

Solution. Orthonormal columns are independent by Proposition 6.2. For $v \in V$, put $w = \sum_i (v^\top v_i) v_i$. Orthogonality gives

$$w^\top w = \sum_i (v^\top v_i)^2, \quad v^\top w = \sum_i (v^\top v_i)^2.$$

The assumed equality now yields

$$\|v - w\|^2 = \|v\|^2 - 2v^\top w + w^\top w = 0.$$

Thus $v = w$ is in the span, proving that the list is a basis. If $r = 0$, the assumption says $\|v\|^2 = 0$ for every $v \in V$; then $V = \{0\}$ and the empty list is its basis.